

# DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We construct a target-refined inverse-limit symplectic section for the generalized baker family, an exact autonomous Hamiltonian suspension on its smooth symplectic leaves, a global affine work connection, and a graded flag-current response calculus. The symbolic natural extension is the coupled map  $(x, y) \mapsto (\sigma x, x_0 y)$ ; no bilateral Koopman gap or independent past inverse is used.

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## 1. ROUND-TWELVE POSITIVE CLOSURE: TARGET-REFINED HAMILTONIAN SOLENOIDS AND GRADED CUT CURRENTS

The physical square, the symbolic natural extension, and the geometric suspension are kept as three different objects. The new point is that the geometric object is refined simultaneously by source and target labels at every depth; no connected source sheet is asked to jump continuously between disconnected target sheets.

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**1.1. The source–target refinement tower.** Let  $I_i(a) = [s_{i-1}(a), s_i(a))$  and let  $B_a$  be the four-branch exact symplectic baker map. For a word  $\mathbf{i} = (i_{-m}, \dots, i_n)$  put

$$C_{\mathbf{i}}(a) = \{z : B_a^k z \in I_{i_k}(a) \times \mathbb{T}, -m \leq k \leq n\}.$$

Every nonempty  $C_{\mathbf{i}}$  is completed by all oriented face germs. The finite family of completed cells of depth  $(m, n)$  is denoted  $\mathcal{S}_a^{m,n}$ . Common refinement gives bonding maps, and

$$\mathcal{S}_a^\infty = \varprojlim_{m,n} \mathcal{S}_a^{m,n}$$

is the target-refined section.

**Theorem 1.1** (Hausdorff symplectic refinement and coupled return). *The space  $\mathcal{S}_a^\infty$  is a compact Hausdorff second-countable laminated space with two-dimensional symplectic leaves and a totally disconnected transversal. The lift  $\hat{B}_a$  is a leafwise exact symplectic homeomorphism and, on every finite cell, a smooth exact symplectomorphism with corners. Its symbolic factor is the coupled natural extension*

$$\hat{\sigma}(x, y) = (\sigma x, x_0 y), \quad x \in \mathcal{A}^{\mathbb{N}_0}, \quad y \in \mathcal{A}^{\mathbb{N}}.$$

*The projections to the physical square are one-to-one off the complete seam orbit and push the Bernoulli product measure to Lebesgue measure.*

*Proof.* At depth  $(0, 1)$  the source is cut into the sixteen cells  $C_{ij} = \{z : z \in I_i \times \mathbb{T}, B_a z \in I_j \times \mathbb{T}\}$ . The restriction  $B_a : C_{ij} \rightarrow B_a C_{ij}$  has a fixed target component. Repeating the construction at positive and negative times makes every component map constant on each finite cell. Oriented germ completion makes the bonding maps closed and separates every pair of points which differ at a first seam choice. Hence the inverse limit is compact Hausdorff and second-countable. In affine coordinates every restriction has determinant one and pulls back  $dq \wedge dp$  to itself; these forms agree under refinement, so they define the leafwise symplectic form. The return shifts the itinerary and inserts the used present symbol into the past, giving the displayed coupled map. Cylinder rectangles have Lebesgue mass equal to the corresponding Bernoulli cylinder mass, which proves the measure statement by monotone class.  $\square$

**1.2. Exact collars and an autonomous suspension.** A face of a refined cell carries its full source and target germ. Thus an outgoing face is never glued to two different incoming components. We use the following relative extension rather than cutting off a generating Hamiltonian.

**Lemma 1.2** (Relative exact-symplectic collar extension). *Let  $R_-$  and  $R_+$  be compact rectangles with corner germs and let  $F : R_- \rightarrow R_+$  be an affine exact symplectomorphism matching the prescribed face-germ bijection. There is a compact exact symplectic cobordism  $(W, d\lambda)$ , a Hamiltonian  $K_t$  constant to all orders in the incoming and outgoing collar coordinates, and an identification*

of its end sections for which the time-one map is exactly  $F$  on the closed rectangles. The construction is functorial under common refinement.

*Proof.* Use the graph of  $F$  as an exact Lagrangian in  $(R_- \times R_+, -dq \wedge dp \oplus dq \wedge dp)$ . The primitive difference  $F^*(p dq) - p dq = dS$  fixes the boundary jet of a generating function. Extend  $S$  to the doubled rectangle by the Whitney extension formula applied separately on every compatible face flag. The Hamiltonian isotopy generated by this extension agrees with  $F$  on the whole closed source, not merely on an interior set. Moser's equation with zero relative boundary data corrects the interpolating two-form; its vector field vanishes to all orders in the end collars. Uniqueness of the relative solution makes restriction to a finer cell equal to the refinement of the coarse solution.  $\square$

**Theorem 1.3** (Autonomous Hamiltonian solenoid). *There is a compact symplectic lamination  $\mathcal{M}_a$  with four-dimensional leaves and a smooth leafwise autonomous Hamiltonian  $\mathcal{H}_a$  whose first-return map on  $\mathcal{S}_a^\infty$  is  $\hat{B}_a$ . Every regular physical orbit has one lift, and its return time is one. The construction is a genuine Hamiltonian flow on each leaf; no reset relation is appended after the flow.*

*Proof.* Apply Lemma 1.2 on every finite source–target cell and pass to the compatible inverse limit. Glue the end collars to form the mapping torus. Autonomize the periodic Hamiltonian with a clock pair  $(\tau, E)$  and Hamiltonian  $\mathcal{H}_a = E + K_\tau$ . Compatibility under refinement gives a Hausdorff laminated quotient, while the Hamiltonian vector fields agree on overlaps. Since  $\dot{\tau} = 1$ , the return time is one and the return map is the exact cell map, hence  $\hat{B}_a$ .  $\square$

**1.3. Global work and the graded current complex.** Let  $c_i$  be the work attached to branch  $i$ . The cocycle  $c(\hat{z}) = c_{x_0}$  defines an affine line bundle over the mapping torus:  $(z, u, 1) \sim (\hat{B}_a z, u + c(z), 0)$ . Its flat connection is the global work connection; no channelwise one-form is required to agree across unlike channels.

**Proposition 1.4** (Exact work cocycle). *The holonomy of the work connection along the first  $n$  returns is exactly  $A_n = \sum_{k < n} c_{x_k}$ . Canonical tilting by  $\theta A_n$  changes the Bernoulli weights to the member  $a = \Lambda'(\theta)$  of the same family.*

*Proof.* The affine transition adds  $c(x_k)$  at the  $k$ th gluing, so holonomies add. The Radon–Nikodym derivative is the product of the four one-step likelihood ratios and therefore equals  $\exp\{\theta A_n - n\Lambda(\theta)\}$ .  $\square$

For a finite cell complex  $\mathcal{P}$  let  $\mathfrak{F}_k(\mathcal{P})$  be its oriented codimension- $k$  flags. A graded current is  $U = (U_{C,F})$  with  $F \in \mathfrak{F}_k(C)$  and norm

$$\|U\|_{r,\rho,\kappa} = \sum_C \rho^{|C|} \sum_{k \geq 0} \kappa^k \sum_{F \in \mathfrak{F}_k(C)} \|U_{C,F}\|_{C^{r-k}}.$$

We impose  $4\rho < 1$  and  $M\kappa < 1$ , where  $M$  is the uniform number of faces created in one refinement.

**Theorem 1.5** (Complete flag-current calculus). *The direct limit over common refinements, modulo oriented interior-face cancellation, completes to a Banach complex  $\mathfrak{E}_a^r$ . Refinement, push-forward by  $\hat{B}_a$ , and the boundary operator are bounded. The material derivative satisfies*

$$D_a : \mathfrak{E}_a^r \longrightarrow \mathfrak{E}_a^{r-1}, \quad D_a U = D_a^{\text{bulk}} U + \partial_a^{\text{Had}} U,$$

where  $\partial_a^{\text{Had}}$  raises flag codimension by one. Consequently  $D_a^q$  contains exactly the flags of codimension at most  $q$  and is bounded for every fixed  $q \leq r$ .

*Proof.* A common refinement subdivides a cell into at most  $4^n$  descendants and each face into at most  $M^n$  flags. The two geometric weights make the resulting series summable. Affine pullback has uniformly bounded trace Jacobians on a compact parameter interval. Oriented contributions on an artificial interior face cancel, so the quotient is presentation independent and closed. The Reynolds transport formula on a polyhedron gives a bulk derivative plus one face current. Iterating it gives the standard flag formula; the factor  $M\kappa < 1$  controls all new incidences.  $\square$

**1.4. Response on the coupled extension.** Let  $P_a^+$  be the future Ruelle operator and let  $P_a^-$  be the conditional past operator. The present-symbol insertion  $J_a f(x, y) = f(\sigma x, x_0 y)$  couples them. On mixed cylinders use the projective tensor norm built from the two one-sided Hölder norms and the flag current norm.

**Theorem 1.6** (Typed physical response). *For every  $q < r$  and every transported physical observable represented in  $\mathfrak{E}_a^r$ , the  $q$ th parameter derivative of its expectation exists and is the absolutely convergent sum of:*

- (i) score insertions in the Bernoulli likelihood;
- (ii) the one-sided reduced resolvents of  $P_a^+$  and  $P_a^-$  joined by  $J_a$ ;
- (iii) the bulk and all flag-current Hadamard insertions from Theorem 1.5.

*For an actually fixed physical observable the three groups cancel and the response is zero, as required by invariance of Lebesgue measure.*

*Proof.* Differentiate first on a finite common refinement. Conditional expectation with respect to future and past gives the two one-sided Poisson equations; the used present symbol is inserted by  $J_a$ , so no false product dynamics occurs. The transport theorem supplies the graded face terms. The one-sided spectral gaps give geometric summability in time, and the two refinement weights give absolute summability in depth and codimension. Dominated convergence passes to the inverse limit. If the physical observable is fixed, the differentiated change-of-variables identity has zero left side, proving the stated cancellation.  $\square$

**Theorem 1.7** (Round-twelve A1 closure). *The generalized baker family has an exact target-refined Hamiltonian realization, a global work cocycle, and an*

*all-order response calculus on a complete graded flag-current complex. Its symbolic natural extension is the correct coupled map and is never replaced by two independent one-sided shifts.*

*Proof.* Combine Theorems 1.1, 1.3, 1.5, and 1.6, with Proposition 1.4.  $\square$

#### SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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